

HOMEWORK 7

Due date: Next Monday.

Exercises: 1, 8, 9, 12, pages 105-107

Exercises: 1, 3, page 111

Exercises: 1, 3, 4, 6, 7, 8, pages 115-116.

You can do Problems 1, 2 and 3 using what you learned from last week.

Problem 1. Given two matrices $A, B \in \text{Mat}_{m \times n}(F)$. Show that $\text{rank}(A+B) \leq \text{rank}(A) + \text{rank}(B)$.

Hint: Translate this problem to a problem on linear maps. You might find the following fact useful: Given two linear maps $T, S : V \rightarrow W$, then $\ker(T) \cap \ker(S) \subset \ker(T+S)$ (check this).

Problem 2. Given two matrices $A, B \in \text{Mat}_{n \times n}(F)$. If $AB = 0$, show that $\text{rank}(A) + \text{rank}(B) \leq n$.

This follows from Sylvester's rank inequality (Problem 3, HW6). But it is a good exercise to prove this directly (without going through the whole proof of Sylvester, because this problem is much easier.) Hint: Think about what can you say about $\text{Im}(T_B)$ and $\text{Ker}(T_A)$. (As usual: T_A denotes the linear map $F^n \rightarrow F^n$ given by $T_A(X) = AX$. Here elements in F^n are viewed as column vectors.)

Problem 3. Let F be a fixed field and n be a positive integer. Denote by $I \in \text{Mat}_{n \times n}(F)$ the identity $n \times n$ matrix. Given $A, B \in \text{Mat}_{n \times n}(F)$.

(1) Show that

$$\text{rank}(A - ABA) = \text{rank}(A) + \text{rank}(I - BA) - n.$$

(2) Show that

$$\text{rank}(I - AB) = \text{rank}(I - BA).$$

Hint: The equation $A - ABA = A(I - BA) = (I - AB)A$ might be useful.

Problem 4. Denote $\alpha = \sqrt[3]{2}$. Consider $F = \{a + b\alpha + c\alpha^2 \mid a, b, c \in \mathbb{Q}\}$. We know that F is a field and it is also a vector space over $\mathbb{Q}$ of dimension 3 from previous HW. We view it as a $\mathbb{Q}$ -vector space.

- (1) Given $x = a + b\alpha + c\alpha^2 \in F$ and the linear map $T_x : F \rightarrow F$ given by $T_x(y) = xy$. It is not hard to see T_x is well-defined and $\mathbb{Q}$ -linear. Here “well-defined” means $T_x(y) \in F$ for $y \in F$. Suppose that $x \neq 0$. Show that T_x is injective and conclude that there exists a $y \in F$ such that $xy = 1$.
- (2) Fix an ordered basis $\mathcal{B}$ of F (as a $\mathbb{Q}$ vector space) and compute the matrix $[T_x]_{\mathcal{B}}$ of T_x with respect to the basis you chose.
- (3) Show that $[T_x]_{\mathcal{B}}$ is invertible (this is similar to part (4) of a problem)
- (4) Do a higher dimensional analogue of this. For example, given a matrix $A \in \text{Mat}_{2 \times 2}(F)$, and consider the linear map $T_A : F^2 \rightarrow F^2$. View F^2 as a $\mathbb{Q}$ -vector space and choose a basis $\mathcal{B}'$ of F^2 over $\mathbb{Q}$. Then compute $[T_A]_{\mathcal{B}'}$ explicitly in terms of entries of A . Show that if A is invertible as an element of $\text{Mat}_{2 \times 2}(F)$ then $[T_A]_{\mathcal{B}}$ is invertible as an element of $\text{Mat}_{6 \times 6}(\mathbb{Q})$.

Exercise 16 page 107 gives a “coordinate free” definition of trace of a square matrix. Please keep in mind this assertion. In Chapter 5, we will see a “coordinate free” definition of determinant. You can find a proof in this [link](#) of Ex 16, page 107. But the proof given there uses heavy computations. In the following problem, we will have a more conceptual proof. Basically, we do Exercise 17, page 107 first and we show Ex 17 implies Ex 16 easily.

Problem 5. Let F be a field and let $V = \text{Mat}_{n \times n}(F)$, which is an F -vector space of dimension n^2 . We consider the trace map $\text{Tr} : V \rightarrow F$. Let W be the subspace of V which is spanned by the matrices of the form $AB - BA$ for $A, B \in V$. Then we know that $W \subset \ker(\text{Tr})$. This space W is exactly the one in Ex 17, page 107.

- (1) Show that $\text{Tr} : V \rightarrow F$ is surjective and conclude that $\dim \ker(\text{Tr}) = n^2 - 1$.
- (2) Show that $\dim W = n^2 - 1$ by explicitly constructing enough linearly independent elements in W . (See the solution of Ex 17, page 107 given in the above link). Conclude that $W = \ker(\text{Tr})$.
- (3) Show that a linear functional $f : V \rightarrow F$ such that $f(AB) = f(BA)$ for all $A, B \in V$ is exactly an element in $\text{Hom}_F(V/W, F)$. Conclude that such an f must be of the form $c\text{Tr}$ for some $c \in F$.

Hint for part (2) if you don't check the solution given online. We expect that $W = \text{Ker}(\text{tr})$. Try to find a natural basis of $\ker(\text{tr})$ (if you still have no idea, try to think about the 2×2 matrices and then 3×3 matrices), and then try to show that they are indeed in W .

Problem 6. Given $V, W \in \text{Vect}_F$ such that $\dim V, \dim W$ are finite. Let $T : W \rightarrow V$ be a linear operator.

- (1) Given a linear functional $f \in W^*$ such that $f|_{\ker(T)} = 0$, show that there exists a linear functional $g \in V^*$ such that $g(T(\alpha)) = f(\alpha)$ for any $\alpha \in W$.
- (2) If T is injective, conclude that T^t is surjective.

Part (1) is a variant/generalization of Ex 12, page 106.

Problem 7. Let V, W be two finite dimensional vector spaces over F . Suppose $\dim_F V = n, \dim_F W = m$.

- (1) Show that the map $\theta : \text{Hom}_F(V, W) \rightarrow \text{Hom}_F(W^*, V^*)$ defined by $\theta(T) = T^t$ is an isomorphism.
- (2) Conclude that there is an isomorphism $\text{Hom}(V, W) \rightarrow (V^*)^m$. Construct this isomorphism explicitly.

(Comment: Part (1) is a generalization of Ex. 7, page 116.)

We can prove the assertion in part (2) of the above problem even we drop the condition that $\dim V$ is finite.

Problem 8. Given two F -vector spaces V, W with $\dim_F W = m$. We don't require $\dim_F V$ is finite. Let $\{\beta_1, \dots, \beta_m\}$ be a basis of W and let $S = \{f_1, \dots, f_m\}$ be the dual basis of W^* . Consider the map

$$\begin{aligned}\theta_S : \text{Hom}(V, W) &\rightarrow (V^*)^m, \\ \theta_S(T) &= (T^t(f_1), \dots, T^t(f_m)).\end{aligned}$$

Show that θ_S is an isomorphism.

(Hint: The proof is not hard.) The above assertion slightly generalizes Ex.6 page 105. Actually Exercise 6 of page 105 gives an inverse map of the one defined above (for a specific choice of S). Try to explain this. Moreover, compare the result in Problem 6 with Problem 1 of HW6.

Let V be a vector space over a field F such that $\dim_F V = n$ is finite. We know that V is isomorphic to V^* since both of them have the same dimension. But this isomorphism is not "canonical". On the other hand, the isomorphism $V \rightarrow V^{**}$ given by $\alpha \mapsto L_\alpha$ is "canonical". The next problem is trying to give you a feeling what the word "canonical" means here.

Problem 9. Let V be a finite dimensional vector space and consider the map

$$\Theta_V : V \rightarrow V^{**}$$

defined by $\Theta_V(\alpha) = L_\alpha$, where $L_\alpha : V^* \rightarrow F$ is defined by $L_\alpha(f) = f(\alpha), f \in V^*$. Let $T : V \rightarrow W$ be a linear map such that both V and W are finite dimensional. Show that the following diagram is

commutative

$$\begin{array}{ccc} V & \xrightarrow{T} & W \\ \downarrow \Theta_V & & \downarrow \Theta_W \\ V^{**} & \xrightarrow{(T^t)^t} & W^{**}. \end{array}$$

Namely, show that $(T^t)^t \circ \Theta_V = \Theta_W \circ T$.

Problem 10. Let F be a field and consider the vector space $V = F^n$ for a positive integer n . An element of V is viewed as a row vector. Let m, k be positive integers.

- (1) For $\alpha = (x_1, \dots, x_n) \in V$, define $f_\alpha \in V^*$ by $f_\alpha((y_1, \dots, y_n)) = x_1 y_1 + \dots + x_n y_n$. The map $f : V \rightarrow V^*, \alpha \mapsto f_\alpha$ is known to be linear. Show that f is an isomorphism. (You don't have to check that $f_\alpha \in V^*$ and $\alpha \mapsto f_\alpha$ is linear).
- (2) Let $A \in \text{Mat}_{m \times n}(F)$ and let $\text{Row}(A) \subset V$ be the row space of A (subspace of V spanned by rows of A). Show that $\alpha \in \text{Ker}(A)$ if and only if $f_\alpha \in \text{Ann}(\text{Row}(A))$. Here $\text{Ker}(A) = \{\alpha \in V : A\alpha = 0\}$ (In the equation $A\alpha = 0$, α is identified with a column vector).
- (3) Let $A \in \text{Mat}_{m \times n}(F)$ and $B \in \text{Mat}_{k \times n}(F)$ be two matrices. Show that $\text{Ker}(A) = \text{Ker}(B)$ if and only if $\text{Row}(A) = \text{Row}(B)$.

Note that part (3) shows that two homogeneous linear systems $AX = 0$ and $BX = 0$ have the same solutions iff they are equivalent. Here “equivalence” is defined in page 4 of the textbook.